

Lab: Explore vs. Exploit -- Why You Do What You Do

Objective

Investigate how **action** serves as a way to fulfill predictions, and explore the tension between **pragmatic action** (getting what you want) and **epistemic action** (learning something new).

Prerequisites

- Completed modules on Systems, Agents, Perception, and Cognition
- Read the module on Pragmatic vs. Epistemic Action

Part 1: The Restaurant Dilemma

Goal: Experience the explore-exploit trade-off firsthand.

Imagine you have five Friday nights to eat out. There are two restaurants: - **Restaurant A:** You have been there before. It is reliably good (7/10 every time). - **Restaurant B:** You have never been. It might be amazing (9/10) or terrible (3/10).

For each Friday, decide where you go and explain your reasoning:

Friday	Choice (A or B)	Why?	What Did You Learn?
1			
2			
3			
4			
5			

1. When did you choose to exploit (stick with what you know)?

2. When did you choose to explore (try something new)?
3. How did earlier choices affect later ones?

Write your response here

Part 2: Actions as Self-Fulfilling Predictions

Goal: Observe how your actions make your predictions come true.

Think of three examples where your behavior caused the world to match your expectations:

Situation	Your Prediction	Your Action	Did the World Conform?
Example: You expect a friend to be annoyed, so you act defensive.	They will be annoyed	Defensive body language	They did get annoyed (because of your defensiveness)
1			
2			
3			

Reflect: How does this connect to the idea that action fulfills predictions rather than just responding to the world?

Write your response here

Part 3: Policy Mapping

Goal: Map out a decision as a set of possible policies.

Pick a real decision you face this week. A "policy" is a sequence of actions, not just a single choice.

1. **Decision:** What are you deciding?
2. **Policy A** (pragmatic -- get what you want): Describe the sequence of actions.
3. **Policy B** (epistemic -- learn something new): Describe the sequence of actions.
4. **Policy C** (a mix): Describe a balanced sequence.
5. Which policy minimizes expected free energy? (Consider both getting rewards and reducing uncertainty.)

Write your response here

Part 4: Action Observation

Goal: Observe someone else's actions through an Active Inference lens.

Watch a person (a classmate, family member, or character in a video) for 5 minutes. Record:

1. What actions did they take?
2. What preferred state were they probably trying to maintain?
3. Were any of their actions epistemic (aimed at gathering information)?
4. Did any action seem to create a self-fulfilling prophecy?

Write your response here

Summary Table

Concept	Definition	Your Example
Pragmatic Action	Action aimed at achieving a preferred outcome	
Epistemic Action	Action aimed at reducing uncertainty / gathering information	
Policy	A planned sequence of actions	
Expected Free Energy	The quantity an agent minimizes when selecting a policy	